

Concept Paper

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Sengtha Chay *

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Concept Paper

Trees, Not Tonnes: Survival-Linked Settlement for Smallholder Green Finance

Sengtha Chay

National University of Management, Phnom Penh, Cambodia; sengtha@num.edu.kh

[†] Disclaimer: This paper describes an independent personal research prototype and design exploration. It is not affiliated with, endorsed by, funded by, or developed for the Royal Government of Cambodia, any ministry, the National Bank of Cambodia, or any other institution. All institutional roles named (licensing registrar, grove authority, commune agriculture officer, sponsor, arbiter, identity registry) are hypothetical placeholders used only to illustrate possible governance structures, and the national payment and identity infrastructure the design assumes does not exist in the form described. Nothing here promises, predicts, or implies adoption by any government or institution. No real grove, plot, licence, programme, beneficiary, or payment is represented. All tokens described are experimental test artifacts with no monetary value, no peg, and no real issuer. **Nothing described is a carbon credit: no component mints, measures, certifies, or represents CO₂, and the argument of this paper is that none should.** The accompanying code is unaudited prototype software provided “as is”.

Abstract

A farmer records the mangroves she planted on the phone she already owns. The record is signed on the device and verifies offline, with no account and no network. She may publish it, and a viewer will draw her grove in three dimensions from what she reported, re-checking each record’s signature on the reader’s own machine and trusting no server for any of it. She may also anchor its hash to a permissioned chain, where a licensed commune agriculture officer who has ridden out and counted what is standing confirms or disputes it. Only that confirmation moves money: an escrowed sponsorship releases funds against the confirmed count, to the grower for the trees still alive and to the officer for the visit that established they were. One record therefore reaches two destinations of very different weight, and what each may carry differs accordingly. A picture and a payment are both *wrappers* around a claim. The picture may be wide — species, counts, photographs, even a carbon estimate — precisely because nothing depends on it; its only obligation is to avoid implying that something does. The payment must be narrow, because something does. The rule governing both is not ours: [Holmström and Milgrom \(1991\)](#) established that the power of an incentive should track the quality of the measure it is attached to, and that attaching a high-powered instrument to a poorly-measured outcome distorts what people do. We apply that result where it has not been applied — to the *representation* rather than the contract — and state it as a rule about artefacts: *a representation may carry no claim stronger than the weakest link in the evidence that produced it, and the correct response to a weak claim is to shrink the wrapper rather than to strengthen the rhetoric*. The voluntary carbon market inverted it, placing a counterfactual about an unobservable future inside a fungible and bridgeable instrument ([Probst et al., 2024](#)), and lost its credibility accordingly. Where consequences begin, therefore, this system records *trees* — a count somebody can walk out and falsify, to which a person holding a licence they can lose has put their name. The carbon figure survives in the diary and in the picture, labelled with the model that produced it; it does not cross onto the chain, and nothing is minted against it. We describe an implementation spanning three public repositories and show that its interesting output is what it refuses: payment is withheld against a photograph older than the promise, against a record nobody has confirmed, and against a confirmation any single licensed verifier has disputed. Token supply falls when trees die, and the registry contains no path by which an issuer may choose it — though, as [Section 5](#) reports, the token’s role hierarchy leaves one open that the registry’s logic does not. The loop this closes runs from the physical world to the digital and back, and its return path is a payment rather than a control signal. By the prevailing taxonomy, which grades digital twins by how automated their data flows are, a farmer photographing her trees makes this a *digital model* — the weakest category available. We accept the classification and dispute the criterion, and propose an accountability axis orthogonal to it: what makes a representation worth acting upon is not how

automatically it updates but who bears the cost of its being wrong. We give a per-layer account of what each mechanism does and does not establish, report a defect and a mis-attribution found in our own system while writing, and are explicit that the oracle problem is neither solved nor solvable here. **This is a design paper and contains no empirical content.** Nothing described has been deployed, no grower has used it, no verifier has been licensed, and the evaluation protocol we specify has not been run. Where we report that code does something, we have executed it; that establishes what the code does and nothing about whether the design works.

Keywords: green finance; payments for ecosystem services; results-based finance; survival-linked settlement; monitoring, reporting and verification; carbon market integrity; smallholder agriculture; mangrove restoration; digital twin; permissioned ledger; tokenization; Cambodia; ICTD

1. Introduction

Everyone has met the tree-planting photograph. A hundred people, a hundred saplings, a press release, and a field of dead sticks eighteen months later that nobody photographs. The failure is not usually insincerity. It is that the money arrives on planting day, so planting is what gets funded, and nothing in the arrangement pays anyone to go back and look.

Going back to look is the expensive part, and it is expensive in a specific and unglamorous way: it is a motorbike ride down a dirt road and an afternoon of somebody's life. Schemes that intend to fund outcomes rather than activities have to pay for that afternoon, and the ones that do not end up trusting a spreadsheet. This is the conditionality problem that the payments-for-ecosystem-services literature has documented for two decades ([Ferraro, 2008](#); [Wunder et al., 2018](#)), and it is not a problem about honesty. It is a problem about who is paid to establish a fact.

1.1. Wrappers and Their Claims

The response of the last fifteen years has been to make the claim tradable. If a tonne of carbon can be represented as an instrument, a market will price it, and the price will fund the verification. The difficulty is what got wrapped. A carbon credit is a claim about a counterfactual — what would have happened absent the project — which is not observable in principle, resting on a measurement that is frequently self-reported, converted by a model with wide confidence intervals. That is among the weakest claims in environmental science, and it was placed inside the strongest available wrapper: fungible, transferable, bridgeable, retirement-eligible. The reckoning that followed ([Probst et al., 2024](#)) was not, we think, a failure of the market's plumbing. It was a mismatch between the strength of the instrument and the strength of the evidence beneath it.

The mismatch is not a new observation, and we do not claim it as one. [Holmström and Milgrom \(1991\)](#) showed that incentive intensity should track the quality of the measure it attaches to; [MacKenzie \(2009\)](#) showed that commensurability in carbon markets is manufactured rather than found, and that deciding what counts as the same unit is a design and political act rather than a discovery; and [Simpson et al. \(2022\)](#) quantified, for biodiversity offsets, how far metric choice drives the outcomes a market produces. What we do is apply those results one layer out, to the artefacts a system shows people, and state the consequence as a rule we then follow:

A representation may carry no claim stronger than the weakest link in the evidence that produced it, and the correct response to a weak claim is to shrink the wrapper rather than to strengthen the rhetoric.

A *wrapper*, in this paper, is anything that presents a claim to somebody else: a token, a payment condition, a rendered picture, a badge — and, as [Section 5.1](#) reports at our own expense, a citation. Wrappers differ enormously in what they assert. A labelled estimate in a garden diary asserts almost nothing. A transferable credit asserts that value has been created, is owned, and may be sold to

someone who will rely on it. The rule says that the second must be built on stronger ground than the first, and that when the ground will not bear it, the answer is a smaller wrapper and not a firmer tone of voice.

1.2. The System

We describe a working system in which a smallholder records the trees in her garden on the phone she already owns, and in which two things may then happen to that record — neither of them automatically, and each governed by the rule.

She may **publish** it, and a viewer will draw her grove in three dimensions from what she reported, re-checking each record's signature and hash on the reader's own device. That picture may be wide. It carries her species, her counts, her photographs and a carbon estimate, and it may carry them because nothing depends on it: no contract reads the picture and no payment consults it. Its whole obligation is to avoid implying otherwise.

She may also **anchor** its hash to a permissioned national chain, where a licensed commune agriculture officer who has ridden out and counted what is standing confirms or disputes it. Here the wrapper narrows at every step. The carbon estimate does not cross; the chain records trees, because a tree is a count somebody can walk out and falsify. Nothing is minted against a count no verifier has confirmed. An escrowed sponsorship releases funds only against a record that is confirmed, undisputed, fresh, and shows enough trees still alive — and it pays the officer for the visit, because unpaid verification is verification that does not happen.

That second path closes a loop from the physical world to the digital and back. The return path is not a control signal but a payment, and what it buys is the continuation of the thing it measures.

1.3. On Being a Digital Model

The prevailing taxonomy grades digital twins by how automated the data flow between physical and digital object is: a *digital model* exchanges data manually, a *digital shadow* adds an automated one-way flow from the physical object to the digital one, and only a *digital twin* integrates both directions, the digital object able to act as a controlling instance of the physical (Kritzinger et al., 2018). By that criterion the system described here is a digital model, the weakest of the three, because a farmer photographing her trees is not an automated data flow.

We accept the classification and dispute the criterion. What makes a representation of a physical thing worth acting upon is not how automatically it updates but whether anyone is accountable for the update. An electricity meter is a poor digital twin and a sound basis for billing, and the reason is a licence somebody can lose rather than a data rate. Section 4 proposes an accountability axis orthogonal to the integration axis, and places this system and others on it.

1.4. Relation to Prior Work by the Author

This paper is the third of a sequence and takes the first two as given.

Chay (2026a) asked where in a supply chain cryptographic key management should sit, and argued that it should be allocated in proportion to institutional capacity — keyless at the origin, keyed in the institutional middle. That paper could leave origin authenticity resting on witnesses and accumulated history, because nothing in it moved when a record was believed.

Chay (2026b) asked how a permissioned national chain may enforce identity below the contract layer so that unmodified applications may deploy on it, and found that such enforcement secures *custody* of a regulated asset without securing *exposure* to it — the remedy lying one layer above the base, at the point where a derivative claim is minted. That paper named the present system among its illustrative applications and included it, in its own words, as evidence that the base layer suffices for real applications rather than as a contribution in itself.

This paper makes the argument that was set aside there. It contributes nothing new to either the record format or the chain architecture, and the contracts it describes are already published. What is

new is the question they answer: not who signed a record, but how large a claim it may be wrapped inside, and what a picture drawn from it is thereby permitted to assert.

1.5. Contributions

- **A known result applied to the representation layer.** The multitask analysis bounds an *incentive*; we bound an *artefact* — a token, a rendered scene, a badge, and, at our own expense, a citation. What a picture may assert is not a question the contract-theory line asks (Sections 3 and 5.1).
- **One record, two destinations of different weight.** The offset-metric literature assumes a single unit for a single market and asks which unit. We let the same observation stay wide where nothing depends on it and narrow where money does, and show the asymmetry is enforced rather than asserted (Section 5.1).
- **Changing the unit rather than the exchange rate.** Simpson et al. (2022) compare metrics inside a market designed to be fungible. We give up cross-project fungibility and choose a unit falsifiable by walking to it, accepting the loss of liquidity as the price (Section 3).
- **An accountability axis** for digital representations of physical things, orthogonal to the integration axis of the prevailing taxonomy, with four levels and a placement of existing systems (Section 4).
- **Payment as the return path.** A description of survival-linked settlement in which the digital state actuates on the physical through money rather than control, and in which the verifier is a named payee (Section 5).
- **The refusals as the demonstration.** A catalogue of what the system declines to do, drawn from the contracts' own view functions, on the argument that a system of this kind is characterised by what it will not assert (Table 3).
- **An operational finding:** that privacy-preserving coarsening of signed data invalidates the signature over it, forcing a three-way choice that we state and cost (Section 5.3).
- **An honest per-layer threat model,** including the explicit non-goal of establishing that any tree exists (Section 6), and a falsifiable evaluation protocol that has not been run (Section 8).

2. Background and Related Work

2.1. Paying for Outcomes, and Paying for Verification

Payments for ecosystem services are conditional by design and conditional with difficulty in practice. Ferraro (2008) gives the theoretical account: conditional environmental contracts are expensive to enforce because the principal cannot observe what the agent knows, and the resulting informational rents fall on exactly the schemes least able to afford them. Wunder et al. (2018) give the comparative one, and their finding is blunt: across a global dataset of operating schemes, the PES-defining principle of conditionality — monitoring compliance and sanctioning detected non-compliance — is seldom implemented, for reasons of administrative ease and competing objectives rather than dishonesty. Angelsen (2017) treats the same question at the level of an international transfer architecture, asking how outcome verification is funded when payment is meant to follow results.

We draw one inference beyond what those papers establish, and mark it as ours rather than theirs: that a scheme unable to fund observation of the outcome will tend to pay instead for the activity it can see, because planting is photographable and survival is not. Neither cited paper makes that claim, and we have not tested it.

The position this paper takes is narrower and does not depend on the inference: the monitoring cost should be a line item inside the settlement rather than an overhead the scheme absorbs, and therefore the verifier should be a named payee in the same instrument that pays the grower. We are not aware of a design that does this, though we would not be surprised to be shown one.

2.2. The Unit, and Who Chose It

The argument of Section 3 has a direct ancestry that we want to state rather than have a reader discover.

Holmström and Milgrom (1991) established the general result: where one outcome is well measured and another is not, attaching a high-powered incentive to the measured one distorts effort away from the rest, and the remedy is a weaker incentive rather than a firmer one. Substituting *instrument for incentive* and *evidence for measure* yields the rule this paper follows, and we claim no novelty for it in that form.

MacKenzie (2009) established the environmental-market form. Commensurability between greenhouse gases is manufactured rather than found — “making things the same” is work that somebody performs — and the choice of what counts as one unit is a design and political act rather than a discovery. That is the argument of Section 3 on unit selection, stated in 2009. Simpson et al. (2022) supply the quantitative version for biodiversity offsets: metric choice drives both ecological and economic outcomes, and a mis-specified exchange rate produces declines in the very thing the market exists to protect.

Where this paper differs is in three places, and we would rather name them than imply a broader claim. The multitask result bounds a *contract*; we bound a *representation*, including one — a rendered picture — that no contract-theoretic treatment considers. The offset literature assumes one unit for one market and asks which unit; we let a single record be wide on a branch where nothing acts on it and narrow on a branch where money does. And where Simpson et al. compare metrics inside a market designed for fungibility, we abandon cross-project fungibility outright and choose a unit falsifiable by walking to it, accepting illiquidity as the price.

2.3. Carbon Market Integrity

Probst et al. (2024) assess achieved emission reductions across crediting project types rather than auditing a single project, which is the level at which a claim about mechanism rather than incidence can be made. We use it for the argument that the failure was structural. A single-project literature also exists and is contested; we have preferred the synthesis.

2.4. What Automation Does and Does Not Remove

Chave et al. (2019) make the argument from inside the remote-sensing community: satellite biomass products are calibrated and validated against field plots, and the field plot does not go away. This is the position this paper takes on digital MRV generally. Where remote sensing can establish a fact it should; where it cannot — and species-level counts in a mixed smallholding are a hard case — the question is not how to automate the measurement but whose licence stands behind the number.

2.5. Tokenised Environmental Assets

Ballesteros-Rodríguez et al. (2024) examine one tokenisation venture empirically. We note plainly that the critical peer-reviewed literature on tokenisation as a design move is thin: what exists is largely industry advocacy or practitioner commentary, and we have not found an assessment we would want to rest an argument on. Our own position — that tokenisation increases the strength of the wrapper, and that the voluntary market’s failures were not caused by insufficiently liquid instruments — is therefore argued in Section 3 rather than cited.

2.6. Digital Twins

The integration taxonomy this paper works against is Kritzinger et al. (2018), discussed in Section 4. Work on authenticity in captured and reconstructed media is relevant to what a viewer may assert and is discussed where it bears, in Section 7.7.

3. Design Principles

Five principles follow from the rule, and every subsequent decision is traceable to one of them.

P1. Match the wrapper to the weakest link.

Where a claim cannot bear an instrument, narrow the instrument. The chain records trees rather than carbon not because carbon is uninteresting but because the estimate cannot carry a transferable asset.

P2. Consequences justify narrowing; their absence permits width.

The same record may be presented widely where nothing depends on it and must be presented narrowly where something does. A display is not held to the standard of a settlement, provided it does not imply that it is.

P3. Pay for the visit.

Verification is somebody's afternoon. A design that does not fund it has decided that it will not happen.

P4. Refuse rather than qualify.

Where a condition is not met, the system declines and says why, rather than proceeding with a caveat. A `canClaim()` that returns false with a reason is worth more to a sponsor than a disclosure that a payment was made on weak evidence.

P5. Every layer is optional and additive.

An unanchored garden is a normal garden and an unpublished one is a private diary. No capability may render a record produced without it invalid.

4. An Accountability Axis

Section 1 conceded that this system is a digital model under the prevailing taxonomy, and asserted that the taxonomy measures the wrong thing. This section makes good on the assertion by proposing what we take to be the missing axis, and by placing the system on it.

4.1. *Why Automation Is the Wrong Grade*

The categorical distinction between digital model, digital shadow and digital twin is drawn on a single criterion: whether data flows between the physical and the digital object are manual, automated in one direction, or automated in both (Kritzinger et al., 2018). It is a good criterion for the question it was built for, which is how tightly a simulation is coupled to a production line, and it is the wrong criterion for whether anyone should act on what the digital object says.

The clearest counter-example is not exotic. An electricity meter is a poor digital twin: it reports one scalar, it is rarely coupled back to anything, and nobody would describe it as a multi-physics simulation. It is nonetheless trusted enough that money moves on its readings every month, in every country, between parties who have never met and do not trust one another. What earns that trust is not the sophistication of the instrument. It is that a licensed party certified the meter, that the licence can be withdrawn, that tampering is an offence, and that a disputed bill has somewhere to go. Meanwhile a modern industrial installation may stream telemetry continuously into a fully integrated digital twin whose readings nobody outside the operator can check, whose calibration nobody certifies, and which reports on the performance of the party that installed it. The second system is far more automated than the first. The first is the one people are willing to be billed by.

The taxonomy has no vocabulary for that difference, because automation of data flow and accountability for data are independent properties. We propose grading the second explicitly.

4.2. *The Axis*

Let a *divergence* be any difference between the state of the physical object and the state the digital object reports of it. Every representation of a physical thing admits divergences; the question an axis

should answer is what happens when one occurs, and to whom. We grade by **the cost borne by the party responsible for a divergence**.

- A0 — Unattributable.** No identified party asserts the state. A divergence has no author and therefore no consequence. Self-reported spreadsheets and unsigned exports sit here, as does any feed whose provenance is lost at the first hop.
- A1 — Attributable.** The assertion is bound to an identity, so a divergence can be traced to a party, but that party has staked nothing on being right. A record signed by a device key belongs here: the signature establishes who said something and never whether it is true, and keys of this kind can be generated by the thousand.
- A2 — Staked.** The assertion is made by a party holding a credential that was obtained at some cost and can be withdrawn. A divergence now costs the responsible party something they cannot cheaply replace. Licensed professionals, certified calibrators and accredited auditors sit here.
- A3 — Adjudicated.** A divergence can be formally contested by another party, and the contest has a defined resolution procedure with a payoff consequence. What distinguishes A3 from A2 is not severity but recourse: somebody other than the asserting party can force the question.

Three properties of this scale are worth stating before it is used. It is *ordinal and not cardinal*: A2 is not twice A1, and the levels are not commensurable across domains. It is a scale of *accountability, not accuracy*: an A3 system can be systematically wrong, and an A0 system can be entirely right; what the scale predicts is who bears the cost of being wrong, which is a different question and, we argue, the one that governs whether a representation can carry consequences. And it grades *the binding, not the instrument*: the same sensor deployed by a party with a licence to lose and by a party with nothing at stake sits at different levels.

4.3. The Two Axes Are Independent

Table 1 places systems on both. The point of the table is the diagonal that is missing: integration and accountability do not travel together, and treating one as a proxy for the other is what allows a highly instrumented and entirely unaccountable representation to be described as state of the art.

Table 1. Integration (Kritzinger et al., 2018) against accountability. The categories are independent, and the cells that matter are the off-diagonal ones.

	Low accountability (A0–A1)	High accountability (A2–A3)
Digital modelmanual exchange	A spreadsheet of self-reported plantings; a photograph in a funder’s report.	The system described here. Manual capture by the grower; a licensed verifier who can be disputed and whose licence can be withdrawn.
Digital shadowautomated one way	A vendor’s telemetry reporting on the vendor’s own equipment, calibrated by nobody and checkable by no one outside.	A utility meter: little integration, a certified instrument, an offence to tamper with, and a defined route for a disputed reading.
Digital twinautomated both ways	A closed-loop installation whose model of itself is asserted by its operator and audited by no one.	A certified control system under a mandatory incident-investigation regime.

4.4. Where This System Sits, and What It Claims

By integration, the system described in this paper is a digital model: a farmer photographing her trees is manual data exchange, and no actuator is attached to anything.

By accountability, the two branches sit in different places, and this is the whole of the design. The published record and the picture drawn from it are **A1**: signed, attributable, staked on nothing — which is exactly why nothing is permitted to depend on them. The settlement path is **A2**, because

verification is accepted only from an address holding a live licence that a registrar can suspend, and **A3** in the part that matters most, because any other licensed verifier can dispute a count and a single dispute withholds the state, with the disputed milestone going to an arbiter who may return the money but may not redirect it.

The claim we wish to make is narrow. For a representation on which money depends, accountability dominates integration: a manual observation somebody can lose a licence over is a sounder basis for payment than an automated stream nobody answers for, and a taxonomy that grades the second above the first will mislead anyone using it to choose. We do not claim the axis is complete, that its levels are the right four, or that it has been validated — it has not; Section 8 sets out what would falsify it. We claim only that the question it asks is the one a funder, a regulator or a grower actually needs answered, and that the existing taxonomy does not ask it.

5. System Architecture

The system spans three public repositories, and the division between them is not incidental: each can be inspected without the others, and each can be operated by a party that does not trust the other two. Figure 1 shows the shape, and the shape is a fork rather than a pipeline.

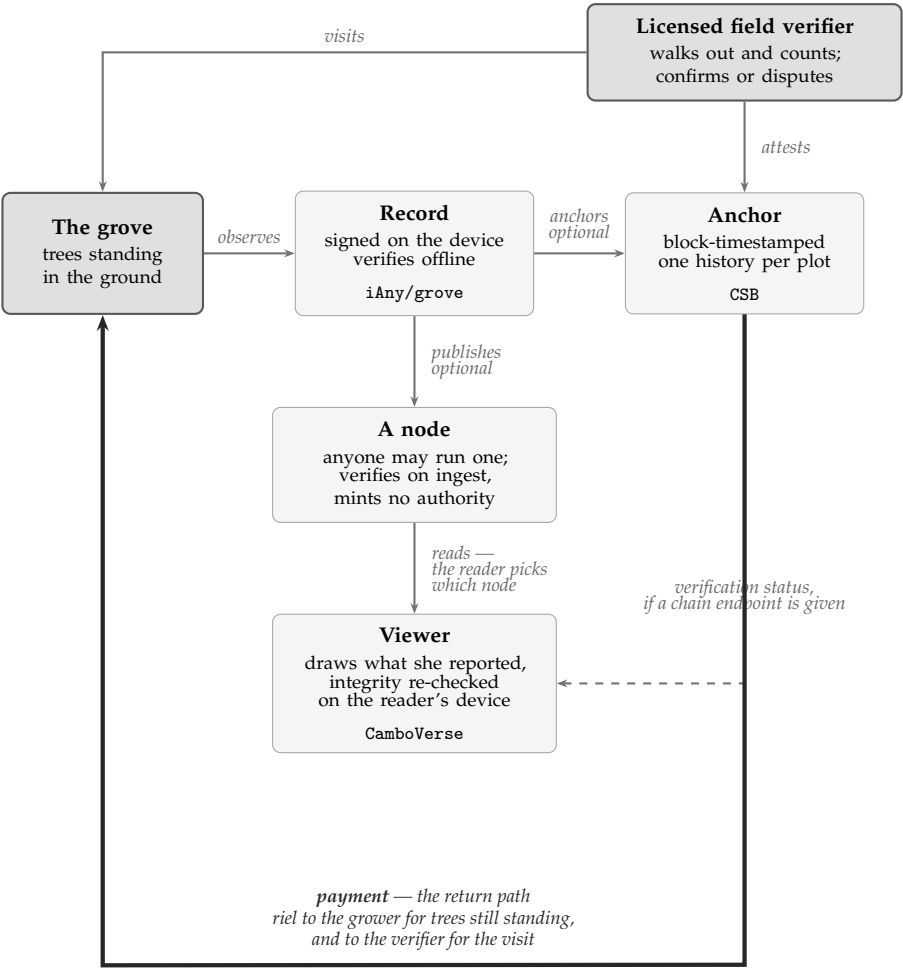


Figure 1. One record, two destinations. The upper path is the consequence path: anchoring gives the record a time its author did not choose and a place for a licensed verifier to sign, and the heavy arrow underneath — a payment, not a control signal — is what makes this a twin rather than a shadow. The lower path is display only. A node is a replaceable mirror that anyone may run — it re-verifies every record it accepts and mints no authority of its own — and a viewer re-verifies again on the reader’s device rather than trusting the node it read from. When a chain endpoint is supplied the viewer additionally shows whether a licensed verifier has confirmed the count, but nothing it draws causes anything to happen. Both destinations are optional: an unanchored garden is a normal garden. Shaded boxes are physical or human; unshaded boxes are software.

A signed record on the grower's phone reaches two destinations. One is a **viewer**, which draws the grove from what she reported, re-checking the record's integrity on the reader's own device and trusting no server for it. Nothing follows from what it draws; no money is attached to any of it, and the viewer is not consulted by anything that pays. The other is the **chain**, which is where the record acquires a time the claimant did not choose and a place for a licensed field verifier to put their name — and it is that confirmation, and only that, against which a sponsorship settles.

The claim of this paper lives in the arrow underneath. A verified count of living trees does not merely describe the grove; it releases money against it, to the grower for the trees still standing and to the verifier for the ride out to count them. The digital state acts on the physical one, and what it buys is the continuation of the thing it measures.

Both destinations are optional. An unanchored garden is a normal garden, and an unpublished one is a private diary. What is not optional is that neither destination may assert more than the record beneath it supports — Section 5.1 sets out what each is permitted to carry.

5.1. What Each Wrapper May Carry

If the rule is that a representation may carry no claim stronger than the weakest link beneath it, then the design question at every boundary is what the next wrapper is permitted to hold. Table 2 answers it for the whole system. It is not a single descent. A signed record forks, and the two branches are governed by the same rule pulling in opposite directions: the display branch may stay wide because nothing depends on it, and the consequence branch must narrow because something does.

The clearest instance is the first boundary on the consequence branch. A record on the grower's phone carries a carbon figure — an estimate produced by applying an allometric model (Chave et al., 2014) to a self-reported measurement — and the application says on its own front page that these are estimates rather than certified credits. The moment the record crosses onto the chain, the unit changes: what is committed is a count of trees, and the anchoring page states plainly that the chain records trees and not carbon. Nothing is corrected at that boundary, because nothing was wrong; a carbon estimate is a perfectly good number for a garden diary. What changes is that consequences begin, and the estimate is not strong enough to carry them.

Two reasons to narrow, at the same boundary.

The first boundary drops four things, and it is worth separating why, because they are not the same argument and a reader will otherwise conflate them. The carbon estimate is dropped on *epistemic* grounds: it is a model applied to a self-reported input, and the chain has no way to check either. The photograph, the coordinates and the plot name are dropped on *privacy* grounds: a permissioned national ledger is readable by everyone permitted onto it, and a farmer's fruit trees are worth stealing. The two rationales happen to coincide here, and the design is stronger for stating that they are independent — a system that narrowed only for privacy would still be free to tokenize an unverifiable estimate, and one that narrowed only for epistemics would publish a map to every grove it recorded.

The estimate survives on the branch that cannot hurt anyone.

The same carbon figure the chain refused appears in the rendered view: a headline total, a per-plant estimate on each label, the allometry named, and the words *never a tradable credit* beneath it. This is not an inconsistency but the argument. A labelled estimate in a viewer that states its own method asserts exactly what it can support, and nothing acts on it. A token asserts transferable value, and a payment asserts that value has been earned. The number did not become more or less true between those rows; the instrument around it did, and the rule is a rule about instruments.

Attribution is a wrapper too.

While preparing this paper we found an instance of the failure it describes in our own interface. The viewer labels its carbon figure with the model that produced it, and the label names a single published allometry (Chave et al., 2014). Only one component of the number comes from there. The

biomass model is that paper's height-inclusive pantropical form, fitted on trunk diameter, total tree height and wood specific gravity. The carbon fraction is an IPCC default, correctly valued and, until this paper, unsourced by anyone who used it — see below. The wood densities are a short in-house table with a constant fallback for unlisted species, where the standard source is a published global database (Zanne et al., 2009; Chave et al., 2009). And where height was not measured, the figure is produced by an in-house height–diameter approximation that appears in no paper at all. The cited work bears on that case twice, and neither reading flatters us: it advises deriving diameter–height relationships from local data wherever possible, which ours are not, and it fits a pantropical model for missing height on wood density, diameter and an environmental variable, which we have not implemented.

Naming one peer-reviewed source for a number assembled from four is the same move as minting a token against a count nobody checked: the strongest available wrapper placed around the weakest link. The difference is only that the wrapper here is credit rather than an instrument, and the rule admits the same two remedies. Either the method is strengthened until it matches the attribution — implementing the published model for missing height, which substitutes an environmental variable for the measurement rather than a rule of thumb — or the attribution is narrowed until it matches the method, naming each component and describing the fallback as the in-house approximation it is. Both are defensible; the present state is neither.

One component took four attempts to source, and the fourth overturned the third.

The carbon fraction is 0.47, called an IPCC default in our source and sourced no further: neither the code nor the repository's history named an edition, volume, chapter or table, and the constant entered in the same commit that carefully attributed the allometry standing beside it.

It is Table 4.3 of Volume 4, Chapter 4 of the 2006 Guidelines, *Carbon fraction of aboveground forest biomass*, at page 4.48 (IPCC, 2006a). And it is the right number. The table gives 0.47 as its overall default, for all domains and all parts of the tree, and gives 0.47 again — with a range of 0.44 to 0.49 — for tropical and subtropical forest, all parts, against which the table lists four sources of its own that we have not gone on to read. Our figure is total aboveground biomass, produced by a pantropical allometry, for trees in a tropical country. The row that applies is the row the constant matches.

We report the route to that finding, because the route is worse than the finding. Two searches reached no primary document and we said so. A third reached Volume 4, Chapter 6 — *Grassland* — which gives 0.47 as the default for *herbaceous* biomass and says it was chosen as more realistic for herbaceous biomass than the previous guideline's value (IPCC, 2006b). From that, and from the fact that Chapter 4's file truncated before its tables, an earlier draft of this paragraph concluded that our constant was a grassland default misapplied to woody fruit trees, and said so at length. That conclusion was drawn from the one chapter we could reach, and it was wrong. The same value sits in both chapters for unrelated reasons, and a partial search finds whichever it can open.

We leave that in rather than quietly replacing it, because it is the failure this paper is about, committed inside the paragraph describing it. Having found an unsourced constant, we reached for the strongest conclusion the available evidence would bear — and the available evidence was one chapter of three, selected by what a tool happened to render. The correct wrapper for a partial read is a partial claim.

What survives is the part that was never about the number. The value was right for the whole time it was unattributed, and nobody in the chain that carried it to our code knew that, including us. A constant that is correct by inheritance is still a constant nobody checked, and the label asserting a provenance is still asserting something its author had not established. Two things follow that the implementation should carry and does not: the table's range, 0.44 to 0.49, which is the honest width of the figure; and the table's structure, which gives 0.49 for tropical wood and 0.43 to 0.47 for foliage, so a later version measuring components separately cannot keep using one number.

And the mechanism is not peculiar to us. A technical guidance document for forest carbon monitoring, written for practitioners, instructs its readers to “use the Carbon Fraction 0.47 t C t⁻¹

dry matter or species-specific values from the literature (per IPCC 2006 GL, V4, Ch4, Table 4.3)” — citing the table for the value without reproducing a single one of its rows ([Winrock International](#)). That instruction is correct. It is also unverifiable by anyone who follows it, which is how the constant reached our source and how it would have reached the next one. Repetition transmits values; it does not transmit the checking, and the reason to go and look was never that the number was likely to be wrong.

The last question closed the same afternoon, and closed better than the absence of a revision would have. Section 4.5 of the 2019 Refinement’s forest chapter lists its tables one by one, marking those it revises (UPDATED) — seven of them — and giving the others a single line. Against Table 4.3 that line reads *No refinement* ([IPCC, 2019](#)). The value is not merely un-superseded by our failure to find a revision; it is un-superseded on the Refinement’s own statement, which is a different and better thing to be able to say.

Table 2. The wrapper ladder. A signed record forks. On the display branch the wrapper may stay wide, because nothing acts on it. On the consequence branch each row carries more weight than the one above and is correspondingly narrower. E marks a narrowing made on epistemic grounds, P one made for privacy.

Wrapper	What it may carry	What it drops, and why	Consequence
<i>The record</i>			
Signed record iAny/grove	Species, measurement, a carbon estimate, the photograph, coordinates, the plot name, the observation date, a device signature.	Nothing. The widest wrapper in the system, and the only one the grower alone controls.	None. A garden diary.
<i>The display branch — nothing reads it</i>			
Rendered view CamboVerse	Everything the published record carries, including the carbon estimate — labelled as an estimate, with its method named and its non-tradability stated — plus, when a chain endpoint is supplied, whether a licensed verifier has confirmed the count.	Nothing on epistemic grounds, because nothing acts on what it draws. It drops only implications: of currency, by a timeline the viewer drives; of vigour, by a deliberately non-photoreal style; of authority, by a source endpoint the viewer may replace.	None. A picture, and a claim about when.
<i>The consequence branch — money depends on it</i>			
Anchored hash GroveAnchor	The record’s content hash, a plot identifier, a count of living trees, a species, the predecessor anchor, a block timestamp, a steward address.	The carbon estimate (E: a model applied to a self-reported input). The photograph, the coordinates, the plot name and the device key (P: this ledger is readable by everyone on it).	None yet. Anchoring is not verification.
Verified count attest()	The same count, with a licensed verifier’s name against it and the licence it was made under.	Self-assertion (E). The grower’s own say-so no longer suffices, and a single dispute withholds the state entirely.	Eligibility. Nothing below can happen without it.
Title shares GroveTitle	One permissioned, transferable share per verified living tree, held only by an address the identity registry admits.	Everything unverified (E), and the registry’s discretion over supply: its only mint path reads the anchored count. Not, however, the token’s own role hierarchy — see the caveat in Section 5.	An asset, inside the identity perimeter.
Payment GrovePledge	Funds released against a record that is anchored, anchored after the milestone opened, confirmed, undisputed, and showing enough trees alive.	Proof older than the promise, and other plots’ records (E). Both are refused by a view a sponsor can read before funding.	Money moves, or does not.

5.2. What Each Leg Contributes

The **record** is taken as given from earlier work (Chay, 2026a): a keyless, content-addressed observation created offline on a commodity phone, whose identifier is the hash of its own contents. It carries no authority. A signature establishes who said something, never whether it is true, and a device key can be generated by the thousand.

The **anchor** supplies the three things a signature cannot. It supplies a time the claimant did not choose, a validator set having agreed the block. It supplies a vouching party with something to lose, since an attestation is accepted only from an address holding a current licence, an active identity attestation, and no enforcement freeze. And it supplies a history that cannot fork, because a new anchor must name the plot's current head as its predecessor. None of this makes a record true. It makes a false record expensive, attributable and singular.

The **node** between them is deliberately weak, and its weakness is the design. Anyone may run one. It verifies every record's hash and signature before storing it, which is what makes it trustless to read from, and it can mint no authority of its own: a record is not more true for being on a node, and a grower who publishes to none loses nothing. Nodes may also be *scoped* — a commune or a province could run one that carries only its own groves — and because a record verifies standalone, a plot appearing on two nodes, or on none, is the same plot. The reference deployment is one pilot node among however many exist; the architecture does not distinguish it from any other.

The choice of node belongs to the reader, not the publisher. The viewer takes a node address as an editable field, so a visitor may point it at a provincial node, at a sceptic's node, or at one they run themselves, and compare. This is what stops the design collapsing into a platform: if availability depended on a single operator, that operator would decide which groves are visible, and the accountability the rest of the paper builds would terminate in a party nobody licensed.

It is worth being exact about what that does and does not prevent, because the two are different. A node *cannot forge*: every record it serves carries a signature and a content hash the reader re-checks, so a node that alters a record produces one that fails verification and is dropped and counted. A node *can omit* — nothing compels it to carry a grove it would rather not show, and a scoped node omits most groves by design. What federation buys is not censorship-resistance but *detectability*: because records verify standalone and a reader may query more than one node, a grove present on one and absent from another is visible to anyone who looks at both. Omission survives; silent omission does not.

The **viewer** is not part of either of those, and its separateness is the point. It reads published records, re-verifies each one's signature and hash on the reader's own device rather than accepting a server's summary, and draws the grove the grower described. Two senses of *verified* meet here and must be kept apart: a record verified on the device is intact and is what its author signed; a count verified in the field is one somebody went and checked. The viewer performs the first for every record it draws, and reports the second only when a chain endpoint is supplied — a check whose result anyone can repeat against the same public state, since a badge in a virtual grove is worth nothing if the only way to check it is to ask the project that drew the badge.

Nothing depends on what the viewer draws. No contract reads it, no payment consults it, and a grower who never publishes loses no entitlement. That is why it may carry the widest claim set in the system — including the carbon estimate the chain declines — and why the discipline it owes is not refusal but disclosure: it must not let a picture imply a currency, a condition, or an authority that the record behind it does not have.

5.3. Coarsening Breaks Signatures

One consequence of publishing signed records is worth reporting on its own, because it is not obvious in advance and it generalises beyond this system.

A record commits to a coordinate. A node redistributing records coarsens coordinates to roughly a kilometre before serving them, because a precise location for a smallholding is somebody's home. Both decisions are right, and they are incompatible: coarsening changes the bytes, so a served record

no longer hashes to the identifier its author signed, and a client that verified the feed as delivered would reject every record in it.

The system resolves this by serving two representations of the same record. The feed carries the coarsened form, which is what map pins are drawn from. A second endpoint returns each record's exact signed bytes by identifier, and it is those that are verified. The client reads identifiers from the feed and re-fetches, six requests at a time, so displaying seven records costs up to eight round trips rather than one.

We report the cost because the trade is general. Any privacy-preserving transformation of signed data invalidates the signature over it, and a system then has three options and no others: serve both forms and pay for the round trips, as here; sign the coarsened form and lose the ability to verify the precise one; or publish precisely and give up the privacy. There is no arrangement in which a reader obtains an exactly verified record while the redistributor withholds the exact data.

The same tension recurs one layer down, in a form we have not yet resolved. A plot identifier is a hash of a plot name, and a name short enough for a verifier to type in a field is short enough to enumerate offline against the chain; salting the commitment closes that, at the cost of making public lookup depend on holding the salt. In both cases privacy is bought with a round trip or a secret. Neither is free, and a design that appears to get it free has usually mislabelled which of the three options it took.

5.4. *The Refusals Are the Demonstration*

A system of this kind is best characterised not by what it permits but by what it declines, since every refusal is a way of getting paid that has been closed. Table 3 lists them by layer. The rows under *Payment* are verbatim returns from `canClaim()`, a view any wallet may call before anyone signs and any sponsor may read before funding.

5.5. *Implementation Status*

The three repositories are not equally developed, and since a reader can clone all three we state the gradient rather than let them find it. The anchoring and settlement layer is the most built out: four contracts, a permissioned deployment, a public read endpoint, and a demonstration script that plays the whole sequence from an empty chain. The record format is a specification with a working offline implementation, carried over from prior work rather than contributed here. The viewer is the newest of the three, and its discipline is of two kinds that should not be confused. Record integrity is enforced by code and observable: a record whose bytes do not match its signed identifier is dropped and counted, and a reader can watch the drop count move. Everything the viewer says about what a picture *means* — that a carbon figure is an estimate, that a scene is dated rather than current, that a source may be substituted — is enforced by labelling rather than by a check that can fail. The first kind can be tested; the second can only be read. That is a real asymmetry in what the two halves of this paper can demonstrate, and we name it rather than let parallel prose imply a symmetry the code does not provide.

A note on evidence.

The two branches are not evidenced in the same way, and the difference should be read before any screen in this paper is taken for a result. Every refusal in Table 3 is exercised by the contract suite's own tests and replayed by a demonstration script that runs from an empty chain, so a reader can reproduce each of them at the archived commit. The viewer's behaviour, by contrast, is observable only in a running instance whose contents are demonstration data — added and removed as the software is developed, unconnected to the settlement testing above, and never intended as a corpus. Nothing displayed there is a measurement. No count shown in it is an adoption figure, a survival rate, or a carbon total for any real grove. Where this paper reproduces such a screen it does so as an illustration of an interface at a stated moment, and we date it accordingly.

A defect found during review, reported rather than repaired.

An independent pass over the contracts, executed against them rather than read, established that the supply guarantee above does not hold as an absolute. The registry's own logic contains no path minting against an unverified count — and says so as intent in a comment — but the title token sets no admin for its minting role, so the office that registers a grove also administers the token and may grant itself that role in one transaction and mint in a second. Worse, the correction is then unavailable: the burn path draws only from the steward, so shares minted elsewhere leave the permissionless re-sync reverting permanently, and the status view reports the cause as shares having been sold on when they were minted from nothing.

Table 3. What the system refuses, and why. Quoted strings are verbatim returns from the contracts’ own view functions.

Layer	The attempt	What the system answers
Anchor	Extend a plot’s history from an address that is not its steward.	Refused. Whoever anchors first owns the chain; appending to a stranger’s garden is otherwise a way to be paid for their trees.
	Anchor a record that does not name the plot’s current head as its predecessor.	Refused. One history per plot, so nobody can quietly keep two and produce whichever is the more flattering.
Attestation	Confirm one’s own record.	Refused. Verification is not self-service.
	Confirm from an address that is unlicensed, suspended, or frozen by an enforcement order.	Refused at three independent gates. A licence alone satisfies only one of them.
	Confirm twice from the same verifier to reach the threshold.	Refused. One attestation per verifier, so a threshold cannot be met by one party repeating itself.
	Outvote a dispute by gathering further confirmations.	Refused. A single dispute withholds verification even against a confirmation: a verifier who says the trees are not there has staked a licence on it, and the answer is a person going to look, not arithmetic.
Payment	Offer last year’s healthy photograph as proof of this year.	<i>“this record predates the milestone: proof of survival has to be newer than the promise”</i>
	Claim against somebody else’s 900-tree grove.	<i>“this observation belongs to a different grove”</i>
	Claim against a fresh record, before anyone has been to look.	<i>“no licensed field verifier has confirmed this record, or it is disputed”</i>
	Settle while either the grower or the verifier is under an enforcement freeze.	The whole settlement reverts — not even the verifier is paid. Compliance is not suspended inside an escrow.
	Have the arbiter of a disputed milestone direct the funds to a third party.	Refused. The arbiter may release to the grower or return to the sponsor, with a recorded reason. A dispute resolver who can name a payee is issuing payment instructions.
Supply	Mint shares against trees no verifier has counted.	Refused <i>by the registry</i> : its only mint path reads the anchored verified count, and the minting role belongs to the registry rather than to any office. Not refused by the token, whose role hierarchy leaves a second path open — a defect, reported below rather than smoothed over.
		Refused in both directions, conditionally. Supply is the difference between the verified count and the tokens outstanding, and the function applying it is open to anyone, so a sceptic may force the correction. But shares minted outside the registry cannot be burned back, and the correction then reverts permanently — so this refusal depends on the one above holding.
	Have the issuer choose the supply, or defer a correction that would reduce it.	
	Treat “nobody has confirmed this yet” as “there are no trees”.	Reverts rather than zeroing a grower’s holding. An unverified new record is absence of evidence.
Viewer	Burn shares that have already been sold on.	Reverts, and says so in words, rather than the registry quietly taking a third party’s property.
		Refused. Every record is re-hashed and its signature re-checked against the signing device’s key, on the reader’s own machine; what fails is dropped, never drawn, and the number dropped is displayed beside the number shown.
	Draw a record whose bytes do not hash to the identifier its author signed.	Refused by construction. The feed is not verifiable — it coarsens coordinates for privacy and so changes the signed bytes — and the client therefore re-fetches each record’s exact signed form by identifier and verifies that instead.
	Accept the redistributing node’s account of what it holds.	

Two consequences for this paper’s argument. The refusal in Table 3 is a refusal by the registry and not by the token, and the row now says so. And the second refusal — that anyone may force the correction — is conditional on the first, which is a weaker claim than we made before the check. We report this rather than repairing the code and quietly restating the invariant, because a paper whose subject is representations that overstate their evidence should not silently improve its own. A repair has since merged, and Section 6 sets out both what it fixes and what it merely relocates; the account here is of the pinned commit and is left as it stood.

We also note two properties of the present design that we regard as unfinished rather than settled. Plot identifiers are hashes of human-chosen plot strings, which is not confidentiality: a low-entropy naming convention is enumerable offline by anyone reading the chain, and the anchored count and species then constitute a public inventory of a grower’s most stealable assets. A salted commitment would close this, at the cost of making public lookup dependent on holding the salt — a tension between public verifiability and the privacy of the party being verified that we do not think admits a clean resolution. Separately, the institutional roles the design depends upon do not exist, and Section 7 treats what the design asks of a state as a first-order limitation rather than as deployment detail.

6. Threat Model

We state per layer what is established and — more usefully — what is not. Table 4 summarises. The organising discipline is that a layer’s entry in the second column is the largest claim it may be wrapped in.

Table 4. What each layer does and does not establish.

Layer	Establishes	Does not establish
Signed record	That a particular device asserted a particular observation, and that the record has not been altered since: its identifier is the hash of its own contents.	That any tree exists. That the device belongs to the grower, or to anyone in particular — keys of this kind can be generated in any number.
Published record and viewer	That every record drawn passed a hash and signature check performed on the reader’s own machine, and that records failing either were dropped and counted.	That the scene corresponds to any physical grove; that the count is right; that the carbon figure is anything but an estimate assembled from a model, a default and a table.
Anchored hash	That this exact record existed no later than a block the claimant did not time, and that the plot has one history, since a new anchor must name the current head.	That the plants are there. Anchoring is not verification, and the system says so at the moment of anchoring.
Attestation	That an address holding a live licence, an active identity attestation and no enforcement freeze confirmed or disputed this record, and which licence it held at the time.	That the verifier visited, counted correctly, or was not paid to say so. A licence raises the cost of a false statement; it does not make one impossible.
Title shares	That supply equals the verified living count at the last synchronisation, and that no path exists by which the registrar could have chosen otherwise.	That the count is true. The token faithfully tracks an attestation, and inherits every weakness of it.
Settlement via the claim path	That funds moved only against a record anchored after the milestone opened, confirmed, undisputed, and showing sufficient living trees — each checkable by a sponsor before funding.	That the trees survived: it establishes that somebody with a licence said so, after the promise was made, and was paid for saying it. <i>And that this is the only path.</i> A second exists, below.
Settlement via the arbiter	That a party holding the arbiter role released a funded milestone, and the reason they gave, which may not be empty.	Everything else. No record is consulted, no window applies, and no proof is required. The role is administered by the council address named below.

Explicit non-goal.

The system does not establish that any tree exists, and cannot. The oracle problem is not solved here. What changes is the cost and the attribution of a false statement: cheating requires a licensed professional to put a registration behind a claim, on a record with a timestamp they cannot backdate, for a payment visible to the sponsor, in a state any other licensed verifier may dispute. That is not proof. It is a worse deal for the fraudster than a spreadsheet, which is the honest claim.

Collusion.

Rural supply chains are dense: the verifier may be a neighbour, a relative, or a member of the same cooperative. Collusion is correspondingly cheaper than in anonymous markets. Detection is also cheaper, because reputation is locally enforced and widely known, and the dispute mechanism gives any other licensed party a way to act on suspicion. We regard the net effect as an open empirical question and one of the more interesting things a field study could measure.

The steward is whoever arrived first.

A plot's chain belongs to the address that anchored it first. This prevents a stranger being paid for another grower's trees, and it creates a converse hazard the design does not currently address: a party who anchors a plot before the grower does holds the chain for that plot name. On a chain where addresses are KYC-verified this is attributable rather than anonymous, and salting the plot identifier makes the name harder to pre-empt, but neither amounts to a claim of ownership over land.

Plot identifiers, and a fix we decided not to ship.

The identifier is an unsalted hash of a human-chosen plot name, so a name short enough for a verifier to type is short enough to enumerate offline against the chain; the anchored count and species then constitute a public inventory of a grower's most stealable assets. That is the defect, and stating it that way understates it. For a grower who publishes, no guessing is required at all: the publishing node's feed emits the plot name verbatim beside the species and the count, and serves a per-plot endpoint keyed on the name. Enumeration is only the residual attack against a grower who anchors and never publishes — which is the case the rest of the system exists to move people out of.

The obvious remedy is a salted commitment, and we report the reason we have not adopted it, because the reason is a better result than the fix would have been. A settlement requires a record anchored *after* the milestone opened, so the identifier must be recomputable for the whole life of a pledge and not merely once. An unguessable identifier is by construction unmemorable, and the plot's stewardship has no setter: lose the secret and the chain cannot be extended, the pledge cannot be proved, and the money returns to the sponsor while the trees stand. The exposure the salt removes is one the grower can already mitigate herself by choosing a name she would not mind a stranger guessing. The exposure it creates is one she cannot mitigate at all, and it falls on the same person.

We therefore ship neither the salt nor a reassurance. The interfaces now state that a short name can be worked out from its hash and advise choosing accordingly. We record this as an unresolved defect rather than a resolved one, and note that the honest form of the fix — carrying the identifier itself in the record and the printed code, so that neither the grower nor any consumer must re-derive it — is a change to a signed format and remains undone.

The root of trust is a single address, and this section omitted it.

Every separation described above is administered from one place. A council address is granted the default administrator role in the constructors of all six contracts — identity, enforcement, the attester registry, the anchor, the title registry and the pledge — and, since a correction described below, on every title token the registry deploys. The offices this section models can each be appointed and replaced by that address, so a reader who takes the licensing registrar, the grove authority, the arbiter and the enforcement holder for four independent parties has read the contracts and missed the constructor.

We name this as an omission in our own threat model rather than as a property of the design, because the design is entitled to a root of trust and a threat model is not entitled to leave it out. Its consequence is inverted from what one might expect: the omission matters least in the deployment where it is most visible — a single-key test network, where every role is that key and nobody is deceived — and most in the separated deployment the architecture is written for, which is the only deployment in which the invariants above mean anything.

Two of the six are not structural remarks but capabilities, and they are the two this paper can least afford to leave implicit. The pledge contract's arbiter role is administered by that address, and the arbiter's release path checks no window, consults no record and requires no proof — only a non-empty reason. So the party administering the system can appoint itself arbiter and settle any funded pledge in full, on the day it is funded, against no trees at all. The attester registry's registrar role is likewise administered by that address, and the registrar issues, suspends and removes field-verifier licences. So the same party can license its own verifiers and suspend inconvenient ones; the contract's refusal of self-attestation raises the cost of that to a second address and no further. The accountable witness on whom the whole settlement argument rests is, in the limit, accountable to the party being witnessed.

Neither is a defect in the sense the others in this section are. An arbiter that cannot release without proof is not an arbiter, and a registrar that cannot withdraw a licence is not a registrar; both powers are the point of the offices that hold them. What was defective was a threat model that described the refusals as the system's guarantee while omitting the address that can decline to be bound by them. The refusals hold against a grower, a verifier and a sponsor. They do not hold against the root, and this paper previously implied otherwise by saying nothing.

A correction since the commits this paper pins makes the point concrete rather than abstract. Review established that the office registering a grove also administered the token created for it, and could therefore grant itself that token's agent role and mint against no verified count, burn a holder's shares, freeze an address or pause the token — and, because the corrective burn draws only from the steward, wedge the supply resynchronisation permanently while its status view blamed the wrong cause. The remedy moved the token's administration to the council: two offices where there was one. It did not remove the capability, and we would rather say so than let a fixed defect read as an absolute guarantee. One bound on it emerged from writing the test rather than from reading the contract: the token gates recipients on the identity registry, so shares minted this way cannot land in an unregistered wallet. Any registered address will do, which makes the bound a real one and a weak one.

Compliance as a threat surface.

The freeze that protects the system from a bad actor is also a power over a grower's payment. A settlement reverts entirely when either party is frozen, which is the correct ordering for a compliance regime and an instrument of pressure in the hands of whoever holds the enforcement role. Section 7 treats the separation of that role from the others as a requirement rather than a deployment detail.

7. Limitations and Future Work

7.1. What This Design Asks of a State

The most consequential limitation is not technical. Every institution this design depends upon is hypothetical. There is no licensing registrar issuing field-verifier credentials, no grove authority, no arbiter, and no national identity registry of the kind the contracts query. The design does not merely run on a test network; it presumes a state that has decided to operate several offices it does not currently operate.

It is worth being precise about what is being asked, because the request is smaller than it first appears and larger in one respect than a technical reader may notice.

Smaller, because the roles are ordinary. A commune agriculture officer is an existing function; licensing agronomists is an existing kind of act; a registry of who held which licence when is an

ordinary administrative record. Nothing here requires a novel institution, only that familiar ones acquire an address and a signing key.

Larger, because the design's central claim rests on those roles being held by *different* parties. If one office licenses verifiers, freezes accounts, issues titles and resolves disputes, then the separation the contracts express is decorative. Following Chay (2026b), which found the same gap in its own deployment, we state plainly that separation of powers here is a property of the contract suite and of a deployment procedure, and not of any instance we have run. A reader should treat every claim in this paper about institutional separation as a claim about what the code permits, not about what any deployment does.

7.2. *There Is No Empirical Content in This Paper*

We state this as a limitation rather than leaving it to be inferred from the absence of results. No grower has used this system. No commune officer has been licensed. No sponsorship has settled against a real grove. No survival rate has been observed, no cost has been recorded, and no person outside the author has tried to complete a task with any part of it.

What we have done is execute the code. The refusals catalogued in Table 3 are exercised by tests and by a demonstration script, and the defect reported in Section 5 was found by running the contracts rather than by reading them. That distinction matters in one direction only: executing code establishes what the code does, and establishes nothing about whether the arrangement it implements would work, would be adopted, or would produce the outcomes it is designed for.

A reader should therefore treat every claim in this paper about *behaviour* as checkable and every claim about *efficacy* as unsupported. Section 8 sets out what would be required to change the second, and the studies it describes remain undone.

7.3. *The Accountability Axis Is Proposed, Not Validated*

Section 4 argues that accountability dominates integration for representations that carry consequences. That is an empirical claim about outcomes and we have not tested it. Its levels may be the wrong four; its ordering may not survive contact with cases we have not considered; and the claim that A2 systems produce fewer costly divergences than A1 systems is exactly the sort of thing that sounds obvious and is sometimes false.

7.4. *Known Defects*

Two properties of the present implementation are defects rather than boundaries, and we would rather list them than let a reader discover them.

Plot identifiers are unsalted hashes of human-chosen plot names and are therefore enumerable, as Section 6 sets out. And the carbon figure's attribution overstates its sources, as Section 5.1 reports. Both have known fixes and neither was fixed at the commits this paper pins.

Corrections have landed since, and we record them rather than revise the account above, because the shape of the correction turned out to be more instructive than the defect.

The first was a single false sentence: that the chain cannot learn a garden's name. The pages now keep the three guarantees that hold — the photograph, the coordinates and the device key never leave the phone — state that the name's hash is public and recoverable for a short everyday name, and advise the grower to choose one she would not mind a stranger guessing. The derivation is untouched, so the enumerability described above stands.

That sentence took six commits to remove. There were eleven instances of it, in eight files across three repositories — a count we give with some embarrassment, having stated it wrongly three times before someone enumerated the removed lines rather than trusting anybody's tally, our own included. The last correction turned on whether a Khmer translation counted: it did not, because at the pinned commit it said only that the hash leaves the phone, which is true. The English beside it said the name never does, which is not. The instances worth naming: it was in an anchoring page's lead paragraph and again in the same page's calldata decoder, beside the value it described; in a garden panel and in

the module header of the file whose own body retracted it forty lines down; in an integrator-facing bridge document; in a viewer's client, its test file's header, and — the sharpest instance — in the *name of a passing test*, which asserted a privacy property the test did not establish. Each pass corrected what it had been shown and searched only the sites it had been given, which is precisely how four instances survived the first four rounds of correction. A test name is the most authoritative claim form a codebase has, being what a later reader cites as having been checked; it was the last instance found and it had been passing throughout.

We draw the lesson rather than the reassurance. When a claim is found to be wrong, the unit of repair is the claim and not the place it was noticed, and the search should precede the first fix rather than follow it. We record that we did not do this, six times, in the course of writing a paper about representations that outrun their evidence.

One change in that sequence went beyond copy. The verification page had taken a plot name typed into a box and accepted one in a query string, putting the name into browser history, referrer headers and server logs on the one page whose purpose is to establish that a record is sound. It now takes a record identifier — a content hash already public on the chain and in every feed — which answers the verifier's question without the name. The old query parameter is refused rather than redirected, on the reasoning this paper applies elsewhere: a fallback that still accepts the weaker input is the path every existing printed code would keep using.

The carbon figure's label was corrected in the same week, and a docs/REFERENCES.md added that records each of the four components and its standing: one published model, one definitional constant, an in-house density table, and one attribution marked UNRESOLVED. The panel now distinguishes the case where height was measured, and names Chave's equation only then; where it was not, it says the approximation is in-house. It prints the word IPCC nowhere, because that attribution was unverified, and the constant's own comment now opens "*do not repeat 'IPCC default' without checking, which is what this comment used to say and what got us here.*"

That attribution has since resolved, and not in the direction we expected. As Section 5.1 reports, 0.47 is the tropical and subtropical value in the table the code should always have named, and it is the right number for what the code does with it. The label was still right to be removed, because at the time it was removed nobody had established that. It may now be restored in a narrower form — naming the table, the domain and the range 0.44 to 0.49 — and it must still not describe the figure as a whole as an IPCC method, since three of its four components are not.

We note what did not change. No number moved and no formula moved. Both defects this section names remain: the plot identifier is still an unsalted hash, and the carbon figure is still assembled from four components of unequal standing. What changed in each case is what the system says about them. What moved is the distance between what the system asserts and what it can support.

That is also why the salted identifier was specified and then declined rather than shipped. The remedy would have made the identifier unguessable and therefore unmemorable, and a settlement requires a fresh record for every milestone, so the secret would be needed for the whole life of a pledge; losing it strands a plot whose stewardship has no setter and lapses a payment against trees that are standing. We could not offer a grower an answer to *what happens if I lose it* that we were willing to write on the screen, and we take that as the finding rather than as a delay. A privacy fix whose failure mode is an unclaimable pledge is itself a wrapper stronger than its evidence, and the rule this paper argues for applies to remedies as readily as to claims.

7.5. What the Display Cannot Demonstrate

The settlement path's discipline is enforced by code and observable: a reader runs the demonstration and watches the refusals. The viewer's integrity checks are likewise observable, since records failing them are dropped and counted. But everything the viewer says about what a picture *means* is enforced by labelling, and labelling cannot fail in a way a reader can watch. That asymmetry is a consequence of the fork — a display nothing depends upon has nothing to withhold — and it means the two halves of this paper are not evidenced equally.

7.6. Scope

The design addresses a single plot with a single steward and a linear history. It does not represent cooperatives that pool plots, land whose tenure is contested, or groves whose stewardship changes hands. It produces no registry-grade certification and is not a substitute for an accredited methodology; it is a cheap measurement and accountability substrate underneath one, and it pays for the trees that are still standing.

7.7. What Photoreal Capture Would Change

The rendering described above is procedural. A count and a species generate a scene, and the scene therefore asserts a count and a species and nothing more; its stylisation is what keeps it from asserting vigour, maturity or health, none of which any field of the record contains. Photoreal capture — Gaussian splatting reconstructed from the grower's own photographs — is planned for the viewer, and it is worth setting down in advance what the rule of this paper implies about it, because the answer is not that nothing changes.

The wrapper is permitted to widen.

A splat is made of photographs that were taken. Its appearance is evidenced in a way a procedural scene's is not, and it may legitimately assert what a grove looked like, because that is what it is a reconstruction of. This is the rule operating in the direction it is usually not asked to: new evidence admits a wider wrapper, exactly as weak evidence requires a narrower one. The estimate that may not become a token may become a picture, and a photograph that was taken may become a scene.

And it acquires a failure mode the drawing does not have.

A photoreal scene implies the present far more forcefully than a stylised one. A stale splat of a grove that has since been cut would be the most persuasive false statement this system is capable of making, and it would be false by implicature while every field in it remained true. Three things therefore become more load-bearing rather than less: the capture date, shown rather than available; the timeline, which makes the scene a claim about a moment instead of a claim about now; and the refusal to derive a count from what the picture appears to show. A reconstruction is evidence of appearance, never of enumeration, and a viewer that let a user count trees in a splat would have reintroduced the self-reported measurement the chain declines.

The record already carries the hook.

A Grove observation commits to its photograph by SHA-256 without publishing the image. A reconstruction built only from photographs whose hashes appear in signed records is one whose inputs are attested: given the record identifiers, a reader can check that every source image was one somebody signed for, on a dated record, in a chain that cannot fork. Work on authenticity in neural and splat-based reconstruction is underway elsewhere — provenance frameworks for captured media, watermarking and steganography for splats, and benchmarks for detecting manipulated reconstructions — and we make no claim to priority. What is particular here is only the terminus of the chain: the source photographs are bound not merely to a camera but to a field observation that a licensed verifier has visited and signed against. Whether that binding survives reconstruction usefully, and what a viewer could check without the images themselves, we do not know, and we note it as the next question rather than as a result.

A constraint the platform imposes.

None of this is unconstrained by the device. The viewer targets low-end Android hardware, because a rendering a grower cannot open on the phone she already owns is not a rendering of her grove in any sense that matters to her. Splat counts, memory ceilings and the uneven availability of modern graphics APIs on such devices bound what can be shown, and that bound is not a detail of implementation: it decides which representations are available at all to the people the system is for.

A photoreal twin that renders only on a workstation would relocate the grove's public existence to people who already have everything.

8. Proposed Evaluation

The studies described in this section have not been conducted, and we make no empirical claims in this paper. We specify them because the claims above are falsifiable and we would rather state how than imply that a working artefact is evidence of efficacy.

8.1. S1: Do the Refusals Hold against an Adversary Who Wants the Money?

Question. The refusals in Table 3 are exercised by tests we wrote. Do they survive attempts by someone trying to get paid without the trees?

Design. An adversarial exercise on a live deployment, with participants given the contract source, a funded pledge, and an incentive to claim it. Instruct them to obtain payment without a qualifying record.

Measures. Successful claims; near-misses and the property that stopped each; refusals whose stated reason a participant found misleading. A refusal that returns the wrong reason is a defect even when it correctly refuses, because a wallet shows the reason to a grower.

8.2. S2: Can a Commune Officer Verify Unaided, and What Does the Visit Cost?

Question. The design assumes a licensed officer can complete a verification with their own wallet, in a field, without support.

Design. A field exercise with agriculture officers who have not seen the system, on real plots, using their own devices.

Measures. Task completion without assistance; time from arrival to confirmation; failures at each of the three gates, and whether the failure message identified the right one. Record the real cost of the visit — travel, hours, opportunity cost — against the fee the pledge pays, since a fee below the true cost reproduces the unpaid-verification failure this design exists to avoid.

8.3. S3: Would a Sponsor Accept It?

Question. Would a funder release money against these guarantees instead of against an audit?

Design. Structured review of the settlement conditions and the public read endpoint by climate funders, corporate buyers and NGO programme staff.

Measures. Which refusals they find reassuring; which guarantees they consider insufficient; whether the ability to read `canClaim()` before funding changes their assessment; what they would require in addition.

8.4. S4: Does Survival-Linked Payment Change Survival?

Question. The whole motivation is that paying on survival funds survival. This is an assertion.

Design. A comparison across plots funded on planting and plots funded on verified survival at twelve and twenty-four months, with survival assessed independently of the payment mechanism.

Measures. Survival rate by funding structure; verification frequency; cost per surviving tree. *We would report a null result.* If survival-linked payment does not improve survival, the design may still be valuable for accountability, but this paper's framing would need to change.

8.5. S5: Does Accountability Dominate Integration?

Question. Section 4 claims that for representations carrying consequences, a manual observation somebody can lose a licence over beats an automated stream nobody answers for.

Design. Compare divergence between reported and independently measured state across representations at different points on both axes — an unaccountable automated feed, a staked manual observation, and a staked automated one — holding the measured quantity constant.

Measures. Magnitude and direction of divergence; whether divergences are random or biased toward the reporting party's interest. The axis predicts that bias tracks accountability level and not integration level. A finding that divergence tracks automation instead would falsify it.

8.6. S6: Does It Render on the Phone People Have?

Question. The viewer targets low-end Android. Does it hold as scenes grow, and would it hold under photoreal capture?

Measures. Frame time, memory and failure rate by device tier and scene size; the point at which labelling degrades; and, for any splat-based successor, whether it renders at all on the target device before it is described as available to growers.

9. Conclusions

The tree-planting photograph is not a story about dishonest people. It is a story about a payment arriving at the wrong moment and no one being funded to go back and look. The response of the last decade was to make the claim tradable, and it failed for a reason worth naming precisely: the weakest claim in environmental science was placed inside the strongest instrument available to carry it.

We have argued for the opposite move. A representation may carry no claim stronger than the weakest link in the evidence that produced it, and where the evidence will not bear an instrument, the answer is a smaller instrument. In the system described here that rule cuts in two directions at once. The picture a grower publishes may be wide — her species, her counts, her photographs, even a carbon estimate — because nothing depends on it and it says so. The payment must be narrow: trees rather than carbon, confirmed rather than asserted, fresh rather than historical, and disputed by any single licensed verifier who says the trees are not there.

The loop that results is a modest one by the standards of the digital-twin literature. A farmer with a phone is not an automated data flow, and by the prevailing taxonomy this is a digital model, the weakest category on offer. We think the taxonomy is grading the wrong property. Nobody trusts an electricity meter because it is sophisticated; they trust it because a licensed party certified it, the licence can be withdrawn, and a disputed reading has somewhere to go. What makes a digital representation worth acting upon is whether somebody is answerable for it, and we have proposed an axis that says so and a design that tries to earn a place on it.

What we have not done is establish that any of it works. No grower has used this, no officer has been licensed, and the institutions the design assumes do not exist. The oracle problem is not solved and cannot be. A licensed officer can be lied to, can be lazy, or can be paid off. What changes is that cheating now requires a professional to put a registration behind a false statement, on a record they cannot backdate, for a payment the sponsor can see, in a state any rival verifier can dispute. That is not proof, and we have tried throughout not to call it one — which is, in the end, the same discipline the paper asks of its own instruments.

Artifact Availability

The system spans three public repositories, released under open licences, each cited at the exact commit this paper describes.

- **Grove records** — the offline record format and the garden application, in the grove/ directory of <https://github.com/sengtha/iAny>, at commit ede2e388. Pilot instance: <https://iany.app/garden>. It is one node among however many are run; the architecture gives it no standing the others lack.
- **Contracts** — AttesterRegistry, GroveAnchor, GroveTitleRegistry with GroveTitle, and GrovePledge — four entries, five .sol files, since each registry deploys its own title — in <https://github.com/sengtha/CSB> at commit 29baa485, together with the grower, verifier and registrar pages and the public GET /grove read endpoint. The wider contract suite of which

these form part is described in Chay (2026b) and archived at <https://doi.org/10.5281/zenodo.21699052> (Chay, 2026).

- **Viewer** — the renderer and its consumption of published records, in <https://github.com/camboversecenter/CamboVerse> at commit 0fde2e9e. Running instance: <https://preview.camboverse.world/>.

The supplement.

A technical supplement, docs/paper-grove-architecture.md in the CSB repository, specifies the architecture at the level of individual files and lines against these same three commits. Every structural claim this paper makes about the system is traceable there to a `repo@commit:path:line` citation and carries an evidence-status marker distinguishing what is specified, what is implemented, what was executed, and what is only argued. The supplement also records which claims have been independently verified and which have not, and reports the defects summarised in Section 5 together with the measurements behind them. A reader who would rather check than take this paper on trust should start there.

Every claim in this paper about the *behaviour of the software* refers to those commits. We note what a commit identifier does and does not guarantee. It fixes content exactly: a reader who can reach the repository obtains precisely the state described, and no later change can alter it. It fixes nothing about availability. Should a repository be removed, made private, or have its history rewritten, the identifier will resolve to nothing, and — unlike a deposit in an archival repository — no independent copy exists. The earlier work these repositories build on is deposited; this one is not, and a reader relying on it in some years should not assume otherwise.

The contents of the live instances are pinned by nothing at all. They hold demonstration data that is added and removed in the course of development, and they are not a dataset: no figure appearing in one of them is a measurement, an adoption count, or a record of any real grove. Screens reproduced in this paper are dated for that reason. Live instances were reachable as of August 2026; the demonstration chain is a private test network and the tokens on it have no monetary value.

Acknowledgments

The virtual grove described in Section 5 is part of CamboVerse, a platform developed by the CamboVerse Center, which the author supervises and to which the author is the principal contributor of the work described here. The Center's wider platform is the work of many hands, and the author thanks its contributors; responsibility for the claims made in this paper about the rendering layer, as for every other claim in it, rests with the author alone.

Disclosure of Generative AI Use

This disclosure is provided in full, and in the form this venue asks for: the use of generative AI and AI-assisted technologies is documented here rather than summarised, and no such tool is listed as an author or credited with any intellectual contribution.

A generative AI assistant (Claude, Anthropic) was used in preparing this manuscript. Its use comprised: locating and summarising related work through web search, and retrieving and checking bibliographic records; reading the three repositories' source, specifications and documentation and summarising their contents; proposing and refining the framing of the central argument, including the reading of the payment path as the return path of a digital twin and the formulation of the claim-and-wrapper rule; drafting the prose of this manuscript; producing the L^AT_EX source, including Figure 1 and Table 3; and assembling the bibliography.

The system this paper describes predates the preparation of this manuscript, and predates it in published form: its contracts and their flow are named and described in Chay (2026b). Its concept, architecture, contract design and every substantive design decision are the author's own. The author additionally discloses that parts of the implementation across the three repositories, and of their

accompanying technical documentation, were written with the assistance of an AI coding tool (Claude Code, Anthropic) working to the author's specifications; the author reviewed, tested and accepted that output and is responsible for the behaviour of the system as described here. No AI tool originated the design it implements.

The research position advanced in this paper — that a representation may carry no claim stronger than the weakest link in the evidence that produced it, and that the correct response to a weak claim is to narrow the instrument rather than to strengthen the language around it — originates with the author, and is visible in the system's own interface copy and in the repositories' commit history well before this manuscript was drafted. Each proposed framing was evaluated by the author and adopted or rejected on its merits.

All AI-generated text has been reviewed, edited and verified by the author, who takes full responsibility for the content of this work, including the accuracy of every claim made about the system and the correctness of every reference.

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